

Supermarket Sales Data Analysis Report

1. Data Overview

The dataset used in this analysis is the “Supermarket Sale” dataset. The dataset contains information about supermarket transactions, including customer type, product category, quantity sold, and total sales amount. The dataset contains approximately 251 rows and 8 columns. The purpose of this analysis is to identify sales patterns and customer purchasing behaviour using descriptive statistics and visualisation techniques.

2. Data Cleaning

During the data cleaning process, missing values in the “Quantity” column were replaced using the average value of the column, which was 11. For missing values in text columns such as “Customer_type” and “Product_category”, mode was used to replace the blank cells. In addition, duplicate rows were identified and removed to improve data accuracy and data quality.

3. Descriptive Statistics and Visualisation

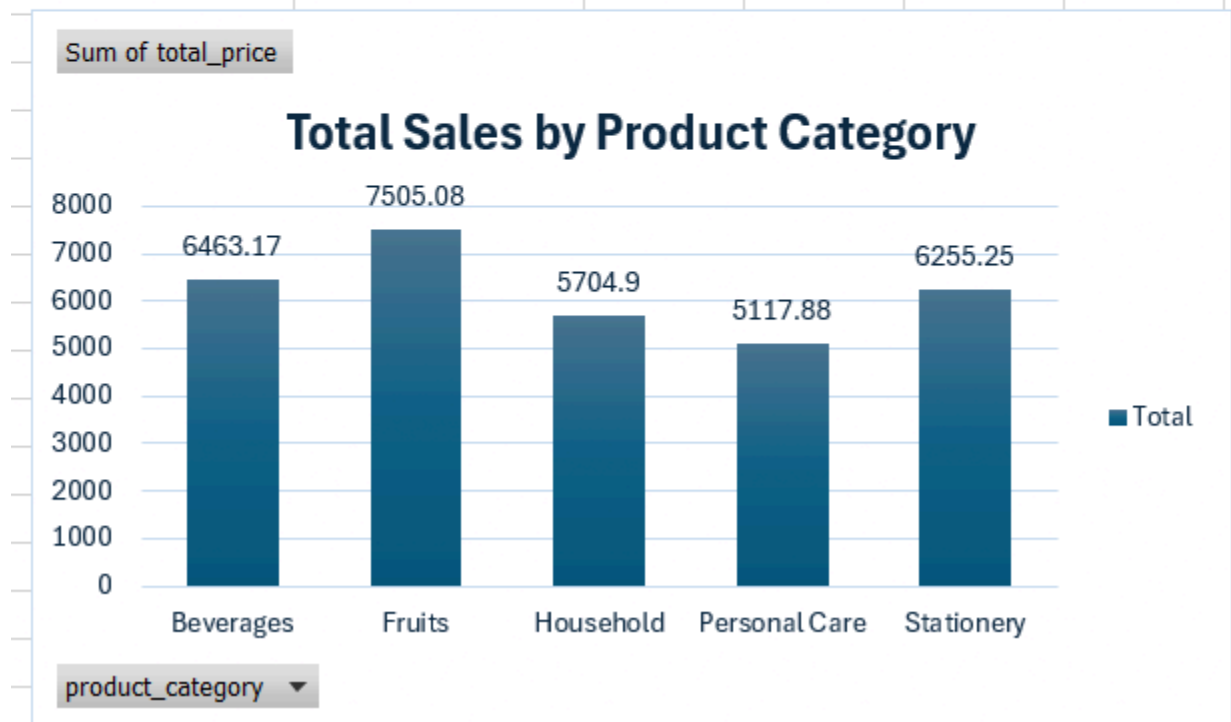
Several visualisations were created to better understand the dataset, including:

- Total Sales by Product Category
- Sales by City
- Customer Type Distribution

Based on descriptive statistics, the average sales amount was approximately 124.19 USD. The highest sales value recorded was 427.14 USD, while the minimum was 2.18 USD. The Fruits category generated the highest overall sales among all product categories.

3.1. Sales by Product Category

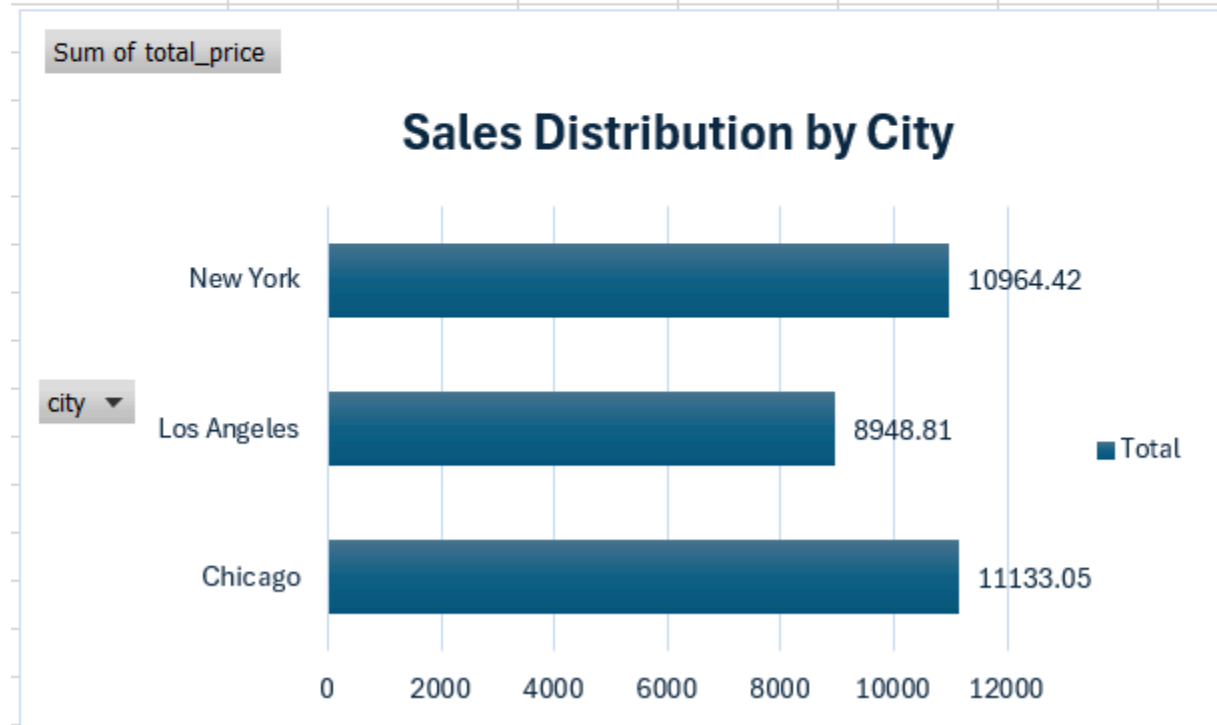
Row Labels	Sum of total_price				
Beverages	6463.17				
Fruits	7505.08				
Household	5704.9				
Personal Care	5117.88				
Stationery	6255.25				
Grand Total	31046.28				



Insight 1: The Fruits category generated the highest total sales among all product categories. This shows that fruit products are highly popular and contribute a large amount to the supermarket's revenue. The supermarket can focus more on this category to increase sales and profits. The supermarket can focus more on this category to increase sales and profits.

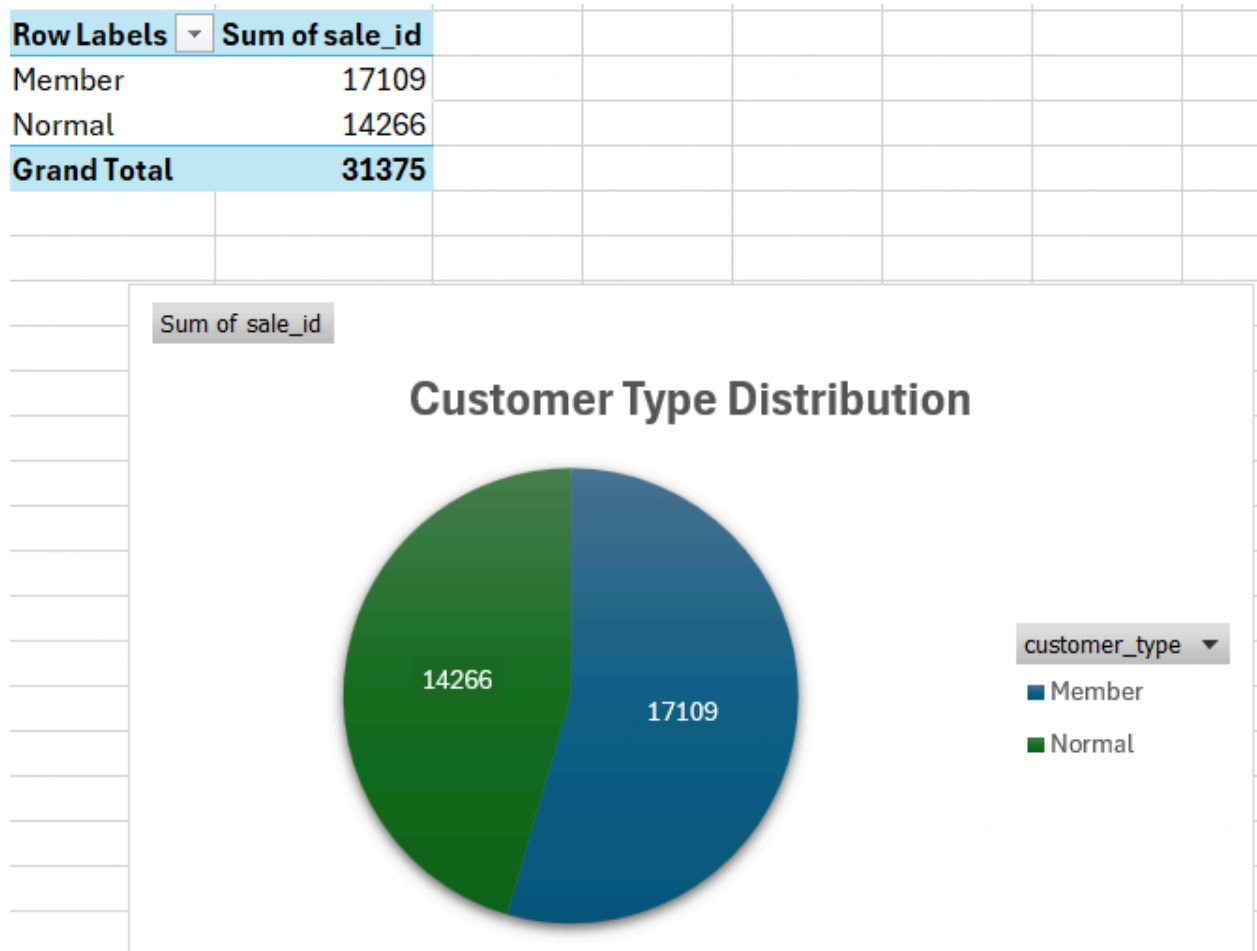
3.2. Sales Distribution by City

Row Labels	Sum of total_price
Chicago	11133.05
Los Angeles	8948.81
New York	10964.42
Grand Total	31046.28



Sales performance was different across cities, which shows that customer purchasing behaviour and demand may vary depending on location.

3.3. Customer Type Distribution



Insight 2: Member customers made more purchases compared to Normal customers. This suggests that membership programs may help increase customer loyalty and encourage customers to shop more frequently.